

Social Media Content & Engagement Analysis Project Report

A Comprehensive Data Analytics Workflow Using Microsoft Excel and Power BI

1. Project Objective & Scope

Project Objective: The primary objective of this project is to analyze social media content, post performance, and user engagement metrics to understand what drives reach, impressions, likes, shares, and follower growth. The project utilizes Microsoft Excel and Power BI to clean, process, analyze, and visualize multi-channel social media data.

Project Scope: The project focuses on handling a dataset of 350+ posts, performing data transformation, establishing calculated measures, and building an interactive 3-page Power BI dashboard. Key attributes analyzed include content types, publishing hours, hashtags, campaign tags, reach, impressions, profile visits, and direct engagement metrics.

2. Tools & Technologies

- **Microsoft Excel:** Used for initial data inspection, formatting, data cleaning, handling missing values, and exploratory analysis of the dataset.
- **Power BI:** Used to build interactive multi-page dashboards, KPIs, dynamic slicers, and advanced visual charts.
- **DAX (Data Analysis Expressions):** Used in Power BI to create calculated columns and measures such as Total Engagement and Engagement Rate.

3. Data Description & Schema

The dataset contains detailed performance logs for individual social media posts. Key fields include:

- **post_id:** Unique identifier for each social media post.
- **date & hour:** Timestamp and posting time to evaluate temporal performance trends.
- **content_type:** Format of the post (e.g., Image, Video, Carousel, Text).
- **Engagement Metrics:** likes, comment, shares, saves, reach, impression.
- **User Action & Metadata:** profile_visits, follow, hashtags, camp_tag (Campaign Tag).

4. Data Cleaning & Preprocessing

Before analysis, the dataset underwent rigorous preparation in Excel:

- **Import & Structural Check:** Imported dataset (~350 rows, multiple columns) into Excel for initial review.
- **Datatype Corrections:** Converted numerical metrics (likes, shares, reach) to proper integers and datetime formats correctly.
- **Removal of Unwanted Columns:** Cleaned redundant tracking fields to optimize data model performance.
- **Missing Value & Outlier Handling:** Checked for blank engagement cells and anomalies, applying standard cleaning steps.

5. Formulas & Measures

New Calculated Measures (DAX):

- 1. **Total Engagement** = SUM(posts[likes]) + SUM(posts[comment]) + SUM(posts[shares]) + SUM(posts[saves])
- 2. **Total Followers Gained** = SUM(posts[follow])
- 3. **Engagement Rate** = DIVIDE([Total Engagement], [Total Reach], 0) * 100
- 4. **Average Reach per Post** = AVERAGE(posts[reach])

6. Power BI Dashboard Architecture

The interactive Power BI report is structured across three core analytical pages:

Dashboard Page	Core KPIs	Visualizations & Layout Structure
Page 1: Overview	Total Reach Total Engagement Total Followers	• Donut Chart: Content Type vs Reach • Line Chart: Engagement Rate by Month • Clustered Column Chart: Content Type vs Likes, Comments, Shares, Saves
Page 2: Content Performance	Total Engagement Total Reach	• Donut Chart: Content Type vs Engagement • Bar Chart: Content Type vs Views • Line Chart: Hourly Post Views Trend • Treemap: Content Type vs Saves
Page 3: Audience & Growth	Profile Visits Followers Gained	• Column Chart: Content Type vs Likes • Treemap: Monthly Follower Growth • Slicer: Content Type Filter • Bar Chart: Content Type vs Shares

7. Conclusion

Project Summary: This project successfully demonstrates an end-to-end data analytics workflow using Microsoft Excel for data preprocessing and Power BI for interactive multi-page dashboards. The resulting insights empower content creators and marketers to optimize posting times, enhance user engagement, and drive follower growth effectively.